# Distrust memo: what in the Westline pack should not be trusted as given

Ritika Raj Parekh · Humaniti AI Build Challenge · 6 November 2026 · Data: TyreSense exports, tyre register, inspections, km log, spec sheet, data dictionary, fleet manager transcript (1 June – 15 July 2026)

| Item | What is wrong, and why it matters | What I did about it |
|---|---|---|
| **Duplicated records** | June and July exports overlap from 18:00 on 30 June to 06:00 on 1 July: 2,869 identical rows appear twice. Left in, that night counts double in every average and alarm count. (F1) | Removed cross-file duplicates before any analysis; a per-file check would have missed them. |
| **Wrong registration** | The Goa truck's July file says KA25B4471 on all 27,082 rows (an A missing). Grouping by the Vehicle column creates a seventh truck. (F2) | Took truck identity from the file name, not the Vehicle column. |
| **Data dictionary** | It labels C as the co-driver (right) side. Tyre serials on the inspection sheets show C = left, D = right in 68 of 70 rows; the 2 exceptions are the documented rotation. Using the label reverses every left-right comparison. (F3) | Used C = left, D = right, verified against serials, not the dictionary. |
| **Register vs transcript: Karwar 5RI** | The register says the new tyre was fitted on 12 July; Ramesh says it was changed at the roadside on 11 July. The sensor was silent for 16 hours, then its battery jumped from 48% to 98%; the export has no sensor ID to explain it. The physical failure time is second-hand ("by two o'clock"). (F12) | Kept both accounts, kept five moments separate and gave no single lead time; question 3 for Ramesh. |
| **Transcript claims** | Accurate on events, less so on details: unsure which truck had the rotation; "the roads are eating the tyres"; Goa alarms "every afternoon"; City truck in the workshop "two Sundays". (F4, F14–F18) | Checked each against the records: rotation was Goa axle 3 on 28 June; hot fills, not roads, explain Hubballi; 13 of 16 trip days; Sundays cannot be confirmed. Anmod 3×/week and the Mangaluru run hold. |
| **Raw TPMS pressures** | Readings move about 0.6–0.7 PSI per °C, so a hot reading against the spec card (a cold value) misleads both ways. The Goa afternoon "alarms" are all 14 tyres cooling together, not leaks; Hubballi fills hot tyres to the cold value. (F15, F17) | Judged pressure against spec only on cold readings (parked ≥3 h), and compared tyres with their twins, which share temperature. |
| **GPS** | Missing on 13–19% of Goa and Karwar records against 1–4% elsewhere. GPS shows that a truck moved, not what the driver did. (F6, F11) | Used GPS for movement and stops only; ignored steps above 120 km/h. GPS distance still matches the km log at 94–98%. |
| **Sensor health** | Four sensors miss about 3× more readings: Gadag 5RO, Goa 4LI, Karwar 3LO, Shivamogga 2RO (battery now 1%). Three of them are also my leak candidates, so a weak sensor and a leaking tyre are not yet separated. (F8, F16, F18) | Flagged rather than filtered (values stay plausible); asked for workshop confirmation before calling them leaks (question 4). |
| **Reporting rate** | The receiver's fast/slow rate looks like an engine signal but is not one; there is no ignition field. (F10, F18) | Used the slow rate only as "parked", confirmed with GPS; made no claim about engine state. |
| **Missing** | No alert log, weather, loads, speed, altitude, ignition, sensor IDs or top-up records. Tread inspections are partial: Goa twice, Shivamogga and Vijayapura never. | Inferred top-ups from pressure steps; called rain "plausible, not shown"; stated each gap as a limit and asked for the alert log first (question 1). |

**What I did trust, after checking:** the km log (no breaks across 270 daily records, and it agrees with GPS); timestamps and GPS fields (internally consistent); and the sensor values themselves (every reading physically plausible). (F7, F8, F11, F18)